Lecture 5 · Part I · Lecture notes

Regularisation and the bias–variance trade-off

Dataset: Titanic, 891 passengers

How to use these notes

The primary text for this lecture is Géron, Chapter 4. These notes are not a replacement for it and do not repeat it: they are the lecture's own argument written out at length — the derivation done slowly, the numbers with the runs that produced them, and the reasoning between one slide and the next that a deck can only gesture at. Read the chapter for the method; read these for why the lecture makes the choices it does. Every figure quoted here is printed by this lecture's notebook, and the course checks that mechanically rather than trusting it.

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1 Two questions the sweep cannot answer

Lecture 4 fitted a logistic regression to 712 passengers and then added polynomial capacity, recording both curves at every degree.

Degree	Columns	Training log loss	Held-out log loss
1	22	0.395	0.468
2	32	0.381	0.467
3	52	0.355	0.538
5	143	0.286	1.922

One curve falls forever; the other turns and climbs. Two questions follow, and they are different questions. *The held-out error is 0.468 and the training error is 0.395 — where does the difference come from?* And *0.468 against an anchor of 0.666 — how far down could it ever go?*

The two have different repairs, and the repairs point in opposite directions. Attack the wrong one and you spend an afternoon making the model worse.

2 Derivation: the bias–variance decomposition

2.1 First, what is random

Not the passenger. Fix one passenger x and ask about them repeatedly. What varies is the *training set* D — you happened to draw these 712 rows — and the *label* y, because two people with identical recorded values did not both survive. Every expectation below is over those two sources and nothing else; the model is deterministic given D.

Symbol	Meaning	Observed?
$\hat{p}_D(x)$	what your model predicts, having seen D	yes
$\bar{p}(x) = \mathbb{E}_D[\hat{p}_D(x)]$	the average prediction over all training sets	no — estimated
$p^*(x)$	the true probability that this passenger survives	never

$y \in \{0, 1\}$ is what actually happened. It is never p^*, and that gap becomes the third term.

2.2 The decomposition

Add and subtract, twice

Hold y fixed and average over D only:

$$(y - \hat{p}_D)^2 = \Big(\underbrace{y - \bar{p}}_{\text{does not depend on } D} + \underbrace{\bar{p} - \hat{p}_D}_{\text{mean 0}} \Big)^2 .$$

The cross term is $2(y - \bar{p})\,\mathbb{E}_D[\bar{p} - \hat{p}_D] = 0$, because $\bar{p}$ is *defined* as $\mathbb{E}_D[\hat{p}_D]$. So

$$\mathbb{E}_D\big[(y - \hat{p}_D)^2\big] = (y - \bar{p})^2 + \underbrace{\mathbb{E}_D\big[(\hat{p}_D - \bar{p})^2\big]}_{\textbf{variance}} .$$

Repeat the trick on $(y - \bar{p})^2$, inserting p^*. Since $\mathbb{E}_y[y] = p^*$ the cross term vanishes again, and $\mathrm{Var}(y) = p^*(1 - p^*)$ because y is Bernoulli:

$$\mathbb{E}_y\big[(y - \bar{p})^2\big] = \underbrace{(p^* - \bar{p})^2}_{\textbf{squared bias}} + \underbrace{p^*(1 - p^*)}_{\textbf{noise}} .$$

$$\mathbb{E}\big[(y - \hat{p}_D)^2\big] = (p^* - \bar{p})^2 + \mathbb{E}_D\big[(\hat{p}_D - \bar{p})^2\big] + p^*(1 - p^*).$$

Term	Means	Cured by
squared bias	wrong on average — the model family cannot reach the truth	more capacity
variance	unstable — a different sample gives a different answer	less capacity, or more data
noise	the label is not a function of the recorded columns	nothing

Generally the identity is $\mathbb{E}[(y - \hat{f})^2] = \text{bias}^2 + \text{variance} + \mathrm{Var}(y \mid x)$; $p^*(1 - p^*)$ is that last term for a Bernoulli label, one instance of the identity rather than the identity itself.

The first two repairs point in opposite directions. That is the whole difficulty, and it is why it is called a trade-off.

Failure condition — an honest caveat before we measure anything

The identity is exact *for squared error*. Our headline metric is log loss, and log loss does not split into three additive terms like this.
So the decomposition is measured on the *Brier score* — the squared error of a probability, which Lecture 4 already reported alongside log loss. The shape of the answer transfers; the numbers are Brier numbers and are labelled as such. Saying "bias–variance" about a log-loss curve is common and slightly wrong. Doing it knowingly, and saying which metric you decomposed, is the difference.

What the derivation buys, before a single number is measured:

The result	What it lets you do
Three additive terms	ask which one is large before choosing a repair
Variance does not mention p^*	estimate it without knowing the truth — refit and look at the spread
Noise depends only on x	measure a floor that no model of these columns can beat
The gap between two curves	read a diagnosis off a plot in ten seconds

The rest of the lecture is that table, executed: measure all three terms on the pipeline of Lecture 4, then repair the one that turned out to be large.

3 Measuring all three terms

3.1 The term you cannot fix

p^* is never observed — but where several passengers share exactly the same recorded values, every model of those columns *must* give them the same number. Whatever their outcomes do inside that cell is a floor.

Grouping on sex, class, age band, family band and port gives 133 distinct cells, of which 102 hold at least two passengers, covering 858 people. Using $k(m-k)/(m(m-1))$, the unbiased estimator of $p(1-p)$ from m Bernoulli draws, the floor is **0.121** Brier — measured, not assumed.

56 of the 102 shared cells are mixed: 657 passengers, 74% of everyone on board, sit in a cell whose answer cannot be certain.

It is an *under*-estimate: two passengers in the same cell may still differ in ways the manifest never recorded. The real floor is at least this high. The cell-by-cell measurement is beyond the syllabus, for context; the decomposition itself is examinable as derived above.

Failure condition — a model measured against zero error

The irreducible noise here is 0.121 Brier, measured from passengers whose recorded values are identical and whose outcomes differ. A model judged against zero rather than against that floor will be tuned forever toward an error that no model of these columns can reach.

3.2 The other two, by resampling

Draw 200 training sets of 400 rows each from the 712 we hold, each one *without* replacement, so no passenger appears twice in a set; fit the pipeline on each, and predict the same 179 test passengers. That is the expectation over D, done by brute force — the only way to see a quantity defined over training sets you did not draw. Bias and noise cannot be separated without p^*, so they are reported as one column, and the floor above is what splits them.

The three measured columns add up to the total at machine precision, at every degree: they are not a *model* of the error, they are the error, rearranged.

Read the first column: 0.130 at degree 1, against a measured noise floor of 0.121. Almost all of it is noise, so the squared bias of a degree-1 logistic model on these features is under 0.01.

There was never much bias to buy back

Every extra degree bought variance and nothing else — the variance term is multiplied by twelve between degree 1 and degree 6 while everything else is nearly flat. That one sentence decides the rest of the lecture.

3.3 Two shapes, two diagnoses

There are only two shapes, and telling them apart takes ten seconds. Both curves plateau high and together: *bias* — the model is too rigid, and more data changes nothing. A gap still open at the last point: *variance* — more data would help, and you may not be able to get any. The repairs are opposites, and applying the left-hand repair to a right-hand curve is the most common wasted afternoon in applied machine learning.

3.4 Learning curves, and a knob we do not have

A learning curve puts *rows* on the x-axis and answers a question the degree sweep cannot: would more data help?

At degree 1 the training and held-out curves have met by the last point; at degree 5 the gap is still open and still falling — that model is starved of rows, not of capacity.

There are 891 passengers on the Titanic

There will never be more. Rows are not a knob we can turn, so the only repair left is to shrink the model.

A *validation* curve puts one hyperparameter on the x-axis instead, and answers which setting is right and whether the optimum is interior. Read the ends first: a minimum sitting on the edge of the range means the range was too narrow, and the answer you read off it is an artefact of where

you stopped looking.

4 Three faults at degree 5

Degree 5 scored 1.922 — almost three times worse than *saying nothing at all*. Not worse than the best model; worse than the anchor of 0.666, worse than a model that has learned only how many people died.

How is that possible when held-out accuracy was poor but not absurd? Because $-\log(0.01) \approx 4.6$: a single passenger given probability 0.01 who then survives contributes 4.6 to the average on its own. Accuracy records that passenger as one error and moves on. Log loss is unbounded above — not a defect, but the metric saying that confident-and-wrong is a different failure from wrong.

4.1 Fault 1 — variance

	Degree 1	**Degree 5**
Columns	22	143
Training rows	712	712
Rows per weight	32	5
Variance term, Brier	0.006	0.068

Five rows per weight. Move one passenger between folds and the fitted surface moves with them. This is the diagnosis the derivation was built to deliver — and two more faults sit underneath it.

4.2 Fault 2 — the minimiser does not exist

If some $\boldsymbol{\theta}$ separates the classes perfectly in the expanded space, then $\sigma(c\,\boldsymbol{\theta}^{\mathsf{T}}\mathbf{x})$ tends to the right label for every row as $c \to \infty$, so the training log loss falls monotonically towards zero as $\|\boldsymbol{\theta}\| \to \infty$. The infimum is never attained.

Degree	1	2	3	4	5	6
Converged	yes	yes	yes	no	no	no
Largest $\lvert\theta_j\rvert$	4.87	5.43	6.32	18.70	11.07	3.98

4.3 Fault 3 — the minimiser is not unique

`FamilySize = SibSp + Parch + 1` exactly, on every row. Standardise the five numeric columns, form $\mathbf{X}^{\mathsf{T}}\mathbf{X}$ and look at its spectrum:

```
>>> np.linalg.eigvalsh(Z.T @ Z)
array([~1e-12,    4.29e+02, 5.43e+02, 7.90e+02, 1.80e+03])  # first: rounding noise
```

The smallest eigenvalue is zero to the precision the arithmetic allows: four columns of information in five columns of data, condition number of order 10^{15} (the exact digits are rounding noise and

vary by machine). A double carries about 10^{16} of relative precision, so roughly one significant digit survives the solve.

5 One term repairs all three

5.1 What adding $\alpha\mathbf{I}$ does to the spectrum

Three lines, and the normal equation is repaired

If $\mathbf{X}^\top\mathbf{X}\mathbf{v} = \lambda\mathbf{v}$ then

$$(\mathbf{X}^\top\mathbf{X} + \alpha\mathbf{I})\mathbf{v} = (\lambda + \alpha)\mathbf{v} :$$

same eigenvectors, every eigenvalue moved up by exactly α. $\mathbf{X}^\top\mathbf{X}$ is positive semi-definite, so $\lambda \geq 0$, so $\lambda + \alpha \geq \alpha > 0$. No eigenvalue is zero, so the matrix is invertible *for every* $\alpha > 0$.

Lecture 3 derived $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top y$ and showed that it fails when $\mathbf{X}^\top\mathbf{X}$ is singular. Those three lines are the repair.

It also bounds the conditioning:

$$\kappa(\mathbf{X}^\top\mathbf{X} + \alpha\mathbf{I}) = \frac{\lambda_{\max} + \alpha}{\lambda_{\min} + \alpha} \leq \frac{\lambda_{\max} + \alpha}{\alpha},$$

and the bound on the right does not mention $\lambda_{\min}$ at all. Whatever the data do — exact collinearity included — the conditioning is controlled by a number you choose. At $\alpha = 1$ the condition number of our singular matrix falls from about 10^{15} to 1798. The dependence is not repaired; it is *dominated*. And κ measures the long, thin valley that slowed gradient descent in Lecture 4: $\mathbf{X}^\top\mathbf{X}$ sets the curvature of the squared-error bowl, and κ is the ratio of its steepest direction to its shallowest. So the penalty is not only a statistical device — it is what makes the optimisation tractable.

Fault	Evidence	A statement about
Variance	a held-out minus training gap still open at degree 5	the estimator
No minimiser	`lbfgs` stops at the iteration cap from degree 4 up	the objective
No unique minimiser	rank 23 of 29; an eigenvalue at 10^{-12}	the design matrix

Three faults, three different objects — and one term repairs all three, the same term in every case.

5.2 The idea, and the three penalties

Stop asking for the best fit. Ask for the best fit *among small models*:

$$J(\boldsymbol{\theta}) = \underbrace{L(\boldsymbol{\theta})}_{\text{fit the data}} + \underbrace{\alpha\,\Omega(\boldsymbol{\theta})}_{\text{stay small}}.$$

Setting $\alpha = 0$ recovers Lecture 4 exactly.

Constraining the model is one of the two cures for variance, and the only one available when the dataset is fixed at 891 rows.

Ridge (also called Tikhonov regularisation) is the ℓ_2 penalty. On squared error,

$$J(\boldsymbol{\theta}) = \mathrm{MSE}(\boldsymbol{\theta}) + \frac{\alpha}{m}\sum_{j=1}^{n}\theta_j^2.$$

The sum starts at $j = 1$: the bias term θ_0 is *not* penalised, because shrinking the intercept would pull every prediction towards zero rather than towards the mean. Equivalently $\Omega = \frac{1}{2}\|\mathbf{w}\|_2^2$, where $\mathbf{w}$ is $\boldsymbol{\theta}$ without the bias.

Lasso (Least Absolute Shrinkage and Selection Operator) is the ℓ_1 penalty,

$$J(\boldsymbol{\theta}) = \mathrm{MSE}(\boldsymbol{\theta}) + 2\alpha\sum_{j=1}^{n}|\theta_j|,$$

and the word *selection* in its name is the point: lasso drives weights to exactly zero, so it performs feature selection as a side effect of fitting. Ridge shrinks everything and eliminates nothing.

Why ℓ_1 produces exact zeros and ℓ_2 does not

Look at the gradient of the penalty near $\theta_j = 0$. For ridge, $\partial(\theta_j^2)/\partial\theta_j = 2\theta_j \to 0$: the push towards zero fades as you approach it, so you arrive only in the limit. For lasso, $\partial|\theta_j|/\partial\theta_j = \pm 1$: the push keeps its size right up to zero, so if the data pulls with less than α the weight reaches zero and stops.
At $\theta_j = 0$ the absolute value is not differentiable; the subgradient is the whole interval $[-1, 1]$, and any data pull inside that interval is held at exactly zero.

5.3 The sweep, on the model that failed

Two penalties — ridge and lasso — swept over `C`, the penalty knob of `LogisticRegression` that Lecture 4 set to 10^6 to switch the penalty off, with ten folds each: two validation curves on one axis.

Both tuned curves sit far below the unregularised 1.922, and the better, lasso, reaches 0.683 — neither reaches the best unregularised degree.

5.4 And now the uncomfortable part

Model	Cross-validated log loss
Degree 5, lasso, tuned	0.683
Degree 1, no penalty at all	0.468
Degree 2, no penalty at all	0.467

The best regularised degree-5 model beats neither of the plain low-degree models we already had — and degrees 1 and 2 differ by 0.001, inside a fold spread of 0.12, so they are tied with each other. Regularisation repaired an over-complex model; it did not turn it into a better one than the simple model — because there was never much bias to buy back. The decomposition said so, and this is what that sentence looks like in a table.

5.5 Ridge shrinks; lasso selects

Refit the degree-5 model across a range of `C` from 10^{-3} to 10^{3} and keep the 143 weights each time; plot each weight against `C` and the two penalties become visibly different objects. The ridge paths approach zero and none arrives; the lasso paths sit at exactly zero and leave one at a time as the penalty weakens, from a handful of non-zero weights at the strong end to nearly all 143 at the weak end.

5.6 Early stopping

Fit iteratively and stop before the optimiser gets where it is going. Descent starts at $\boldsymbol{\theta} = \mathbf{0}$ — the smallest possible model — and each epoch moves the weights outward, so held-out error falls, reaches a minimum, and then rises. No penalty term and no extra term in the objective: just a stopping rule, whose knob is the epoch count.

6 Choosing α, and the test set once

Failure condition — a hyperparameter chosen on the rows you then report

Looping over 17 values of `C`, scoring each on the test set and keeping the best, produces a minimum over seventeen *noisy* estimates — and the minimum of noisy estimates is biased downward even when every estimate is individually unbiased.
The code never writes to the test set. It reads it seventeen times, and reading is enough.

Measured over 20 splits, against the honest version that picks `C` by 5-fold cross-validation inside the training part:

Reported number	Log loss	Spread over 20 splits
Chosen on the test set, scored on the test set	0.451	0.049
Chosen by cross-validation, scored on the test set	0.470	0.045

Optimism of 0.019 of log loss, about 4.1%. On 13 splits of 20 the first number is the flattering one, and the two procedures picked the same `C` on 7 splits of 20.

The procedure without the problem is `GridSearchCV`: same seventeen candidates, same runtime, *two* numbers instead of one. Report both. The second is allowed to be worse than the first, and if it is much worse that is itself the finding. Nothing derived from the test set may appear in the training path — including the choice of any hyperparameter.

6.1 The honest numbers

179 passengers, untouched since the split. Five candidates, all of whose hyperparameters were fixed by cross-validation before they were scored.

Model	Log loss	Brier	Accuracy
Degree 5, no penalty — where the sweep ended	1.727	0.189	74.9%
Degree 5, ridge, tuned	0.804	0.173	77.1%
Degree 5, lasso, tuned	0.677	0.244	65.9%
Degree 1, scikit-learn defaults	0.433	0.134	82.7%
Degree 2, no penalty — the winner	0.427	0.134	82.7%

The winner improves on the degree-5 model Lecture 4 finished with by 1.300 of log loss. And row four deserves as much attention: `LogisticRegression()` with every default left alone — degree 1, `C=1.0`, `penalty="l2"` — scores 0.433, statistically the same model as the winner.

The result nobody wanted

A whole lecture of penalties, and the best system on the test set is a two-line model anyone could have written before Lecture 4 started.
Report it anyway. The alternative is choosing the interesting model over the better one.

7 The notebook

What to change

In the penalty sweep, widen the saga grid `CS_SHORT` to `np.logspace(-8, 4, 25)`. Does the lasso minimum move? If it does, the number in §5.3 was an artefact of where we stopped looking — read the ends of a validation curve first.

8 Exercises

The five set at the end of this lecture's deck, in the style of the written paper. The solutions are on Lecture 6's deck and in the exercise book.

1. Write the bias–variance decomposition of the expected squared error, and name the one term no model can reduce. [4]

2. A learning curve shows training and validation error both high and close together, and flat as data is added. Diagnose it, and say whether more data would help. [5]

3. Ridge adds αI to $X^{\mathsf{T}}X$ before inverting. Give the numerical consequence, in terms of the condition number. [4]

4. Lasso sets some weights to exactly zero; ridge shrinks every weight but sets none to zero. Explain the difference from the gradient of each penalty near $\theta_j = 0$. [4]

5. You tune α over a grid of seventeen values by scoring each on the test set, then report the best test score as your estimate of future performance. Name the error, and quantify its direction. [5]

Summary

Expected squared error splits exactly into squared bias, variance and irreducible noise — over the training set and the label, never over the test point. The gap between a training curve and a held-out curve *is* the variance term; curves plateauing together mean bias instead. On this data the noise floor is measurable at 0.121 Brier, from passengers whose recorded values are identical, and the squared bias of a degree-1 model is under 0.01 — so every extra degree bought variance and nothing else.

Degree 5 failed three ways at once: high variance at five rows per weight, no minimiser under separation, and no unique minimiser under an exact collinearity whose smallest eigenvalue was of order 10^{-12}. Adding $\alpha\mathbf{I}$ makes $\mathbf{X}^\top\mathbf{X}$ invertible for every $\alpha > 0$ — the normal equation of Lecture 3 repaired in three lines — and bounds the condition number by a quantity you choose.

Ridge shrinks, lasso selects, early stopping keeps the weights small by not arriving. None of them beat choosing the right capacity: the winner on the test set was degree 2 with no penalty at 0.427, against 0.677 for the best tuned degree-5 model. A penalty is a repair, not an upgrade. And a hyperparameter chosen on the test set cost 0.019 of log loss here, one-sided, growing with the size of the search.